

Karthik Rachamolla

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EDUCATION

University of Cincinnati

Bachelor of Science in Computer Science, Minor in Finance

Anticipated Graduation: May 2028

GPA: 3.90

EXPERIENCE

ATech Training

May 2025 – Aug 2025

Embedded Software Engineer Intern

Walton, KY

- Built a full-stack app with React and Python that interacts with devices using the Web Bluetooth API
- Designed a vehicle diagnostic interface using the ESP32 microcontroller family, enabling real-time data retrieval and display and accelerating vehicle diagnostics by 40%

Motz Engineering

Aug 2024 – Dec 2024

Software Developer Intern (Database Systems)

Cincinnati, OH

- Engineered a C# Project Manager integrated with the Ajera API, reducing data retrieval times by 60% using optimized SQL queries and benefiting 50+ employees daily
- Designed and implemented a second custom full-stack internal project management tool, enabling detailed customization of checklists, client guidelines, and designs while improving project tracking and reducing time spent searching for client information
- Led a significant overhaul of an internal database management tool, improving efficiency, refining code organization, and implementing features aligned with employee needs

PROJECTS

the-search-thing | Search / Information Retrieval

Jan 2026 – Present

- Designed and built a local-first semantic search desktop application using Rust, Electron, and Helix DB to unify search across multiple content types, targeting sub-millisecond query latency for fast local file discovery
- Developed a multimodal indexing pipeline that chunks video, summarizes images, and generates vector embeddings to enable semantic search over large local media collections

ivy | Agentic AI

Feb 2026 – Mar 2026

- Built an AI manuscript analysis platform using Python and Azure AI to process book-length PDFs, automatically extracting narrative structure and detecting plot inconsistencies, reducing manual continuity review from months to minutes
- Architected a scalable Agentic pipeline that transforms unstructured manuscripts into summaries and timelines, enabling accurate analysis of narratives exceeding 1000+ pages

Quartz | AI Security & Privacy

Nov 2025

- Designed a Chrome extension using JavaScript and Cloudflare Workers that performs real-time prompt scanning across LLM web apps, detecting and anonymizing 8+ classes of sensitive data to reduce PII leakage in AI workflows
- Built a prompt security pipeline with regex entity detection, reducing manual prompt sanitization effort by approximately 80%

SKILLS

Languages: Rust, Python, JavaScript/TypeScript, C++

DB Systems: SQL (Postgres, MySQL), NoSQL (MongoDB, Azure Cosmos, Gremlin DB)

AI/ML: LLM pipelines, embeddings, NLP, CNNs

Tools: Azure AI, React, Electron, Cloudflare, Git